

The actual final bootstrap in Weil II

The proof of the curve theorem in *Weil II*, 3.2.3–3.2.15, uses a more specific mechanism than tensor amplification alone. A local calculation makes the weights of every relevant nonconstant constituent integral. An earlier analytic theorem excludes weight 2 and all larger weights. Consequently these integral weights are at most 1. The remaining constituents are geometrically constant on the projective line, where their first cohomology vanishes. These two exclusions make a product-square argument improve a universal error e to $e/2$. The proof below retains the schemes, coefficient sheaves, boundary terms, maps, twists, and indices which make that improvement possible.

The primary source is Pierre Deligne, *La conjecture de Weil II*, Publications mathématiques de l’IHÉS 52 (1980), 137–252: the introduction, printed pp. 140–141; 1.8.4, pp. 175–176; 2.2.10, p. 196; 3.1, pp. 197–200; 3.2, pp. 200–204; and 3.3, pp. 204–207. All numbered results cited in this section are Deligne’s published results unless a displayed tag begins with WDB. The local current S20 English and French texts were read alongside the controlling published French scan. This section is an exposition of that proof and its exact algebraic steps, not a claim that a construction over the absolute base has already acquired the finite-field sheaves in this proof.

Conventions, original weight, and the universal assertion

Fix a finite field $\mathbb{F}_q$, an algebraic closure $\overline{\mathbb{F}}_q$, a prime $\ell \neq \text{char } \mathbb{F}_q$, and Deligne’s coefficient isomorphism $\iota : \overline{\mathbb{Q}}_\ell \rightarrow \mathbb{C}$. For a scheme or sheaf with subscript 0, omission of that subscript means extension to $\overline{\mathbb{F}}_q$. Use geometric Frobenius and the weight convention

$$w_q(\alpha) = 2 \log_q |\iota \alpha|, \quad w_{q^a}(\alpha^d) = w_q(\alpha), \quad w_q(\alpha^r) = r w_q(\alpha). \quad (\text{WDB1})$$

A Tate twist (a) multiplies Frobenius eigenvalues by q^{-a} and lowers their weights by $2a$.

Let U_0 be a smooth curve and let $\mathcal{F}_0$ be a lisse pointwise ι -pure sheaf of weight β on U_0 . Deligne proves the universal assertions

$$B_k : \quad w_q(\alpha) \leq \beta + 1 + 2^{-k} \quad \text{for every eigenvalue of } F \text{ on } H_c^1(U, \mathcal{F}), \quad k = 0, 1, 2, \dots \quad (\text{WDB2})$$

The quantifiers range over all such curves, sheaves, weights, and finite fields. This universality is used when the induction is applied to new coefficient sheaves on the base of a pencil, rather than only to the original $\mathcal{F}_0$.

Here is the precise relation to the weight-zero formulation of 3.2.4. Choose $c \in \overline{\mathbb{Q}}_\ell^*$ with $\iota(c) = q^{-\beta/2}$, and let $\mathcal{L}_c$ be the constant geometric Weil sheaf whose arithmetic datum is Frobenius c . Deligne’s twisting operation 1.2.7 gives

$$\mathcal{F}_0^{[0]} = \mathcal{F}_0 \otimes \mathcal{L}_c, \quad H_c^1(U, \mathcal{F}^{[0]}) = H_c^1(U, \mathcal{F}) \otimes \mathcal{L}_c, \quad \alpha^{[0]} = c\alpha, \quad w_q(\alpha^{[0]}) = w_q(\alpha) - \beta. \quad (\text{WDB3})$$

Thus the original weight and eigenvalue are recovered by tensoring with $\mathcal{L}_{c^{-1}}$ and multiplying by c^{-1} . We use $\mathcal{F}_0^{[0]}$, not a change in the meaning of $\mathcal{F}_0$, for the auxiliary weight-zero proof. This twist is made in the Weil-sheaf setting allowed by Deligne; it is not an assertion that every such character extends to an ℓ -adically continuous character of the profinite arithmetic Galois group.

We will repeatedly use the following elementary linear algebra. If a linear automorphism preserves a finite filtration, a basis adapted to the filtration makes its matrix block triangular. Its characteristic polynomial is the product of the characteristic polynomials on the successive quotients. Hence a uniform upper weight bound is inherited by subspaces, quotients, and finite extensions. For a convergent Frobenius-equivariant spectral sequence, each term on a later page is a subquotient of an earlier term; the abutment has a filtration with the E_∞ terms as successive quotients. These facts prove the weight comparisons below without assuming spectral-sequence degeneration or Frobenius semisimplicity.

The two earlier inputs: a strict bound and local integral shifts

Deligne's 1.8.2 supplies the initial estimate $w_q(\alpha) \leq \beta + 2$, hence B_0 . For a smooth affine curve it applies directly with the open immersion in 1.8.2 taken to be the identity: the Euler product and the H_c^2 denominator have no zero in $|t| < q^{-(\beta+2)/2}$, so the numerator has none there. For a projective component, remove a closed point and use the surjection from the compactly supported first cohomology of the remaining affine curve, proved explicitly in (WDB7)–(WDB8) below. Connected components are handled by a finite extension and (WDB1).

The stronger earlier result used in the induction is 2.2.10:

$$\alpha \in \text{Spec}(F; H_c^1(U, \mathcal{F})) \implies w_q(\alpha) < \beta + 2. \quad (\text{WDB4})$$

The inequality is strict. Deligne proves it in Section 2 using the Hadamard–de la Vallée-Poussin method and the cohomological expression for the relevant L -function. After weight-zero twisting and reduction to an irreducible representation, 2.2.9 excludes zeros on the boundary line $\text{Re } s = 1$; 2.2.8 and 2.2.9.1 identify this with the absence of the corresponding numerator roots on $|t| = q^{-1}$. The possible geometrically constant character is accounted for by the simple pole stated there, rather than discarded. Its pole order equals the order of the H_c^2 denominator zero, so a numerator zero at that point would contradict the exact simple pole and is excluded as well. Since a reciprocal root $t = \iota(\alpha)^{-1}$ has modulus $q^{-w_q(\alpha)/2}$, the absence of numerator roots for $|t| \leq q^{-1}$ is exactly $w_q(\alpha) < 2$ in weight zero. This is an earlier proved input in the order of the paper; it is not deduced by assuming the conclusion of 3.2.3.

The second earlier result is local monodromy purity, 1.8.4. If a lisse sheaf on a punctured smooth curve is pointwise ι -pure of weight b , and M is the centered local monodromy filtration on its geometric generic stalk V , then

$$\text{Gr}_j^M V \text{ is } \iota\text{-pure of weight } b + j, \quad j \in \mathbb{Z}. \quad (\text{WDB5})$$

In particular every Frobenius eigenvalue of the local Weil representation has weight in $b + \mathbb{Z}$. Under unipotent inertia the graded pieces have trivial inertia: writing $\rho(\sigma) = \exp(t_\ell(\sigma)N)$, the operator N lowers M by two, so its exponential acts as the identity on every Gr_j^M .

The weight shift in (WDB5) is itself proved in 1.8.4, not read off from the existence of a nilpotent operator. To recall its numerical content, twist to weight zero and use a finite covering so that all inertia is unipotent. Then $\ker N = V^I$, and 1.8.1 says its eigenvalues have modulus at most 1. For a primitive bottom piece $P_{-j}V$, Deligne's tensor decomposition 1.6.14.4 makes

$$P_{-j}V \otimes P_{-j}V(-j) \text{ a direct summand of } P_0(V \otimes V). \quad (\text{WDB6})$$

If α occurs in $P_{-j}V$, the resulting eigenvalue $\alpha^2 q^j$ therefore has modulus at most 1, giving $|\iota\alpha| \leq q^{-j/2}$. Applying the same argument to the dual primitive piece, whose corresponding eigenvalue is $\alpha^{-1} q^{-j}$ by 1.6.14.5, gives $|\iota\alpha| \geq q^{-j/2}$. Equality follows. The primitive decomposition 1.6.14.3 and the Tate shifts on its strings give (WDB5). Thus the integral shifts later used by the pencil argument have already been tied to the actual local Weil action.

The exact open-extension map used by the induction

Let $j : W_0 \hookrightarrow C_0$ be a dense open immersion of curves, with finite complement, and let $\mathcal{K}_0$ be a constructible sheaf on C_0 . Extension by zero gives the short exact sequence

$$0 \longrightarrow j_! j^* \mathcal{K}_0 \longrightarrow \mathcal{K}_0 \longrightarrow \mathcal{K}_0 / j_! j^* \mathcal{K}_0 \longrightarrow 0. \quad (\text{WDB7})$$

The quotient is supported on finitely many points. Over the algebraic closure its positive-degree compactly supported cohomology vanishes: each geometric point has only degree-zero cohomology

and a finite disjoint union has the direct sum of those cohomologies. The long exact sequence of (WDB7) consequently gives the Frobenius-equivariant surjection

$$H_c^1(W, j^* \mathcal{K}) \twoheadrightarrow H_c^1(C, \mathcal{K}). \quad (\text{WDB8})$$

If W_0 is smooth and $j^* \mathcal{K}_0$ is lisse and pointwise pure of weight b , assertion B_k applied to the source proves that every eigenvalue of the target has weight at most $b + 1 + 2^{-k}$. This is precisely 3.2.5(k). No purity of the possibly singular boundary stalks of $\mathcal{K}_0$ is assumed in this step.

The product and pencil, with all three reduction hypotheses

Assume B_k . Initially impose the three hypotheses of 3.2.6 on the auxiliary sheaf $\mathcal{F}_0^{[0]}$.

- (A) If X_0 is the smooth projective compactification of U_0 , the local monodromy of $\mathcal{F}^{[0]}$ at $X - U$ is tame and unipotent.
- (B) Put $S_0 = X_0 \times X_0$, $V_0 = U_0 \times U_0$, and $D_0 = S_0 - V_0$. Choose a projective embedding and a hyperplane pencil with axis A_0 satisfying 3.1.1(A)–(D): the axis is transverse to S and disjoint from D ; the only singularities of the pencil on S are ordinary quadratic points outside D ; its only singularities on the normalization of D are ordinary quadratic points away from the preimages of the nodes of D ; and each exceptional fibre contains exactly one exceptional point.
- (C) These exceptional points are defined over $\mathbb{F}_q$.

Write $P_0 = A_0^* \simeq \mathbb{P}_{\mathbb{F}_q}^1$ for the parameter line of the pencil, and use the incidence surface

$$\tilde{S}_0 = \{(s, t) \in S_0 \times P_0 : s \in H_t\}, \quad \pi : \tilde{S}_0 \longrightarrow S_0, \quad \bar{f} : \tilde{S}_0 \longrightarrow P_0. \quad (\text{WDB9})$$

It is the blow-up at the finite set $S_0 \cap A_0$. Put $\tilde{V}_0 = \pi^{-1}(V_0)$, let $\tilde{j} : \tilde{V}_0 \hookrightarrow \tilde{S}_0$ be the open immersion, and set $f = \tilde{f}\tilde{j}$. The actual coefficient sheaf is

$$\mathcal{G}_0 = \text{pr}_1^* \mathcal{F}_0^{[0]} \otimes \text{pr}_2^* \mathcal{F}_0^{[0]} \quad \text{on } V_0. \quad (\text{WDB10})$$

Both tensor factors are retained. In terms of the original sheaf this is $(\mathcal{F}_0 \boxtimes \mathcal{F}_0) \otimes \mathcal{L}_{c^2}$, so its original weight 2β has been recorded by an explicit invertible twist. Its auxiliary weight is zero.

Let $T_0 \subset P_0$ be the finite set of exceptional values, $W_0 = P_0 - T_0$, and $w : W_0 \hookrightarrow P_0$. Define

$$\mathcal{R}_0^b = R^b f_! \pi^* \mathcal{G}_0 = R^b \bar{f}_* \tilde{j}_! \pi^* \mathcal{G}_0. \quad (\text{WDB11})$$

These sheaves are lisse over W_0 . The compact-support Leray spectral sequence is Frobenius-equivariant:

$$E_2^{ab} = H^a(P, \mathcal{R}^b) \implies H_c^{a+b}(\tilde{V}, \pi^* \mathcal{G}). \quad (\text{WDB12})$$

Since both the base and the fibres have dimension one, total degree two has only the terms $(a, b) = (0, 2), (1, 1), (2, 0)$.

Why the middle coefficient sheaf has pure constituents

Let $x \in |W_0|$. Denote the proper fibre of $\bar{f}$ by $\tilde{S}_x$, its open part by Y_x , and the open immersion by $j_x : Y_x \hookrightarrow \tilde{S}_x$. Proper base change applied to (WDB11) gives the exact stalk identification

$$(\mathcal{R}_0^b)_{\bar{x}} \simeq H^b(\tilde{S}_{\bar{x}}, j_{x!} \pi^* \mathcal{G}) \simeq H_c^b(Y_{\bar{x}}, \pi^* \mathcal{G}). \quad (\text{WDB13})$$

These are identifications of representations of geometric Frobenius over $\kappa(x)$. The open fibre is a smooth curve for $x \in W_0$; it need not be connected.

There is a notation defect in the published display 3.2.7.1, printed p. 202: it writes ordinary $H^1(Y, \pi^* \mathcal{H})$ for an open fibre while the left side is $R^1 f_!$. The controlling French scan and both current S20 transcriptions show the same display. Equation (WDB13) states the corrected support convention and its base-change proof. Ordinary cohomology of the open fibre is not used in place of compactly supported cohomology.

Set $\mathcal{H}_0 = \mathcal{G}_0 \oplus \mathcal{G}_0^\vee$. At a closed point the eigenvalues of $\mathcal{G}_0$ have ι -modulus one. The images of the eigenvalues of its dual are their inverses, hence their complex conjugates. Therefore the characteristic polynomial of the direct sum has real coefficients, and $\mathcal{H}_0$ is pointwise pure of weight zero and ι -real. Deligne's 3.2.1–3.2.2 applied to the smooth fibre in (WDB13) shows that $w^* R^1 f_! \pi^* \mathcal{H}_0$ is ι -real. The sheaf $w^* \mathcal{R}_0^1$ is an actual direct summand of it because pullback and $R^1 f_!$ preserve finite direct sums. The purity criterion 1.5.1 now gives a finite filtration of $w^* \mathcal{R}_0^1$ whose successive quotients are lisse and pointwise ι -pure. This proves 3.2.7 with its coefficient and support maps specified.

To make this use of 3.2.1 explicit, let $\mathcal{E}_0$ be pointwise ι -pure of weight b and ι -real on a smooth geometrically connected curve C_0 . Put $P_a(t) = \iota \det(1 - Ft, H_c^a(C, \mathcal{E}))$. Grothendieck's formula says that $P_1/(P_0 P_2)$ has real Taylor coefficients and hence is a real rational function. The roots of P_2 all have modulus $q^{-(b+2)/2}$ by 1.4.3; those of P_0 , if any, have modulus $q^{-b/2}$; and (WDB4) excludes roots of P_1 on the former circle. Consequently the poles on that circle, with their multiplicities, are exactly the roots of P_2 . They are stable under complex conjugation, so P_2 has real coefficients and constant term one. If C is affine, $P_0 = 1$. If it is projective, the correct scalar duality is

$$H^0(C, \mathcal{E})^\vee \simeq H_c^2(C, \mathcal{E}^\vee(1)). \quad (\text{WDB13a})$$

The coefficient sheaf on the right is again ι -real and pure, so the just-proved degree-two argument and inversion of eigenvalues show that P_0 is real. It follows that P_1 is real. If geometric components are permuted in a cycle of length d , their cohomological characteristic polynomial is $\det(1 - F^d t^d, H_c^a(C', \mathcal{E}))$ for one component C' ; the preceding argument over $\mathbb{F}_{q^d}$ makes this polynomial real, proving the extension 3.2.2.

There is a second printed notation defect in the proof of 3.2.1, p. 200: it writes $\mathcal{E}^\vee(-1)$ in place of the $\mathcal{E}^\vee(1)$ in (WDB13a). The printed group equals $H^0(C, \mathcal{E})^\vee(-2)$, so it still implies the required reality after the real scalar factor q^2 is retained. Formula (WDB13a) records the exact untwisted duality used here; the later 3.2.15 itself has the correct (1) twist.

Vanishing cycles detect the nonconstant weights

The calculation in 3.1 treats exactly three exceptional geometries: an ordinary node of the fibre away from D , tangency of the fibre to one smooth branch of D , and passage of the fibre through a node of D . Let $\bar{t} \in T$ and $\bar{x}$ be its unique exceptional point. The vanishing-cycle sheaves are supported at $\bar{x}$ and vanish in every degree other than one. If $\varepsilon(B)$ is the one-dimensional sign representation of the two branches or two local sheets, the calculation is

$$\begin{aligned} \text{node outside } D : \quad & \Phi_{\bar{x}}^1 = \mathcal{G}_{\bar{x}}(-1) \otimes \varepsilon(B), \\ \text{tangency to } D : \quad & \text{Gr}^a \Phi_{\bar{x}}^1 = \text{Gr}^a \mathcal{G}_{\bar{x}}^{\text{loc}} \otimes \varepsilon(B), \\ \text{node of } D : \quad & \text{Gr}^a \Phi_{\bar{x}}^1 = \text{Gr}^a \mathcal{G}_{\bar{x}}^{\text{loc}} \otimes \varepsilon(B). \end{aligned} \quad (\text{WDB14})$$

Here the latter two expressions mean the stalks of the constant extensions of the local graded sheaves of 3.1.2; they do not mean that the original sheaf was defined at a deleted boundary point. For the first case the Picard–Lefschetz vanishing cycle supplies the twist (-1) . For tangency, apply vanishing cycles to $0 \rightarrow j_! \bar{\mathbb{Q}}_\ell \rightarrow \bar{\mathbb{Q}}_\ell \rightarrow \bar{\mathbb{Q}}_{\ell D} \rightarrow 0$: the constant ambient term has no vanishing cycles, which shifts the divisor computation by one. At a node of D , the corresponding two-step

exact sequence through the normalization of D shifts the computation by two and gives the same degree-one sign representation. These are 3.1.3.1–3.1.5.4.

For each original boundary point of $X - U$, (WDB5) applied to $\mathcal{F}_0^{[0]}$ gives integral local weights. Hypothesis (A) makes its graded sheaves locally constant across the puncture. Taking their external tensor product gives the local filtration of $\mathcal{G}_0$ used in (WDB14); its weights are sums of integers. At an interior point the weight is zero. The sign representation has Frobenius eigenvalue 1 or -1 and weight zero, and (-1) adds two to the weight. Every case of (WDB14) therefore proves

$$\text{all Frobenius eigenvalues of } \Phi_x^1 \text{ have integral weights.} \quad (\text{WDB15})$$

The proof retains the distinction between the three geometric supports while obtaining the same integrality conclusion from them.

Let $\bar{\eta}$ be the local geometric generic point at $\bar{t}$. The vanishing of Φ^a for $a \neq 1$ gives the exact sequence 3.2.9.1

$$0 \rightarrow \mathcal{R}_{\bar{t}}^1 \rightarrow \mathcal{R}_{\bar{\eta}}^1 \rightarrow \Phi_{\bar{x}}^1 \rightarrow \mathcal{R}_{\bar{t}}^2 \rightarrow \mathcal{R}_{\bar{\eta}}^2 \rightarrow 0. \quad (\text{WDB16})$$

In particular $\mathcal{R}^1$ has no sections supported at $\bar{t}$. Varying $\bar{t}$ gives the injective sheaf map $\mathcal{R}_0^1 \hookrightarrow w_* w^* \mathcal{R}_0^1$ of 3.2.8.

Choose the finite pure-constituent filtration G on $w^* \mathcal{R}_0^1$ obtained above and extend it by intersection:

$$G^a \mathcal{R}_0^1 = \mathcal{R}_0^1 \cap w_* G^a(w^* \mathcal{R}_0^1), \quad \mathcal{K}_{a,0} = \text{Gr}_G^a \mathcal{R}_0^1. \quad (\text{WDB17})$$

The intersection is in $w_* w^* \mathcal{R}_0^1$. Write b_a for the pure weight of $w^* \mathcal{K}_{a,0}$. At $\bar{t}$, this is the filtration induced by intersection with the subspace in (WDB16); consequently taking its associated graded sequence gives

$$0 \rightarrow (\mathcal{K}_{a,0})_{\bar{t}} \rightarrow (\mathcal{K}_{a,0})_{\bar{\eta}} \rightarrow A_{a,\bar{t}} \rightarrow 0, \quad A_{a,\bar{t}} \text{ a subquotient of } \Phi_{\bar{x}}^1. \quad (\text{WDB18})$$

For completeness, this subquotient statement is elementary for an inclusion $E \hookrightarrow V$ and a filtration of V : filter E by $E \cap G^a V$ and V/E by the images of $G^a V$. The associated graded sequence is exact, so the cokernel in degree a is a graded piece of V/E . Here V/E injects into Φ_x^1 by (WDB16).

If $A_{a,\bar{t}} \neq 0$, select any one of its Frobenius eigenvalues λ . As a subquotient of the vanishing cycle it has an integral weight by (WDB15). As a quotient of the local Weil stalk of the pure sheaf $w^* \mathcal{K}_{a,0}$, its weight lies in $b_a + \mathbb{Z}$ by (WDB5). Thus

$$A_{a,\bar{t}} \neq 0 \implies w_{N(t)}(\lambda) \in \mathbb{Z} \cap (b_a + \mathbb{Z}) \implies b_a \in \mathbb{Z}. \quad (\text{WDB19})$$

If instead every $A_{a,\bar{t}}$ vanishes, (WDB18) identifies the special stalk with the entire generic stalk at every boundary point. The image of specialization is inertia invariant, so inertia acts trivially and the sheaf is lisse across every such point. It is therefore lisse on $P \simeq \mathbb{P}^1$; its geometric monodromy is trivial because $\mathbb{P}^1$ over an algebraically closed field is simply connected. It follows that $\mathcal{K}_a$ is geometrically constant. Frobenius on its constant fibre can still be nontrivial. We have proved the exact alternative of 3.2.9:

$$\mathcal{K}_a \text{ is geometrically constant on } P, \quad \text{or} \quad b_a \in \mathbb{Z}. \quad (\text{WDB20})$$

The pure weight b_a can be measured at any closed point of W_0 . There its eigenvalues are among those of (WDB13) with $b = 1$. Applying the strict bound (WDB4) to the weight-zero coefficient sheaf on that smooth fibre proves $b_a < 2$, which is 3.2.10. Combining this with (WDB20) gives

$$\mathcal{K}_a \text{ not geometrically constant} \implies b_a \in \mathbb{Z}, \quad b_a < 2, \implies b_a \leq 1. \quad (\text{WDB21})$$

This is the decisive return to the earlier exclusion. A weak bound $b_a \leq 2$ would leave the integer 2 available and would not start the improvement from $k = 0$.

Leray transfers the surviving error only once

If $\mathcal{K}_a$ is geometrically constant with fibre L_a , then

$$H^1(P, \mathcal{K}_a) = H^1(\mathbb{P}^1, \overline{\mathbb{Q}}_\ell) \otimes L_a = 0. \quad (\text{WDB22})$$

The equality is Frobenius-equivariant and holds for every Frobenius action on L_a . No numerical weight bound for L_a is required. In the other case, (WDB8), B_k , and (WDB21) give

$$\alpha \in \text{Spec}(F; H^1(P, \mathcal{K}_a)) \implies w_q(\alpha) \leq b_a + 1 + 2^{-k} \leq 2 + 2^{-k}. \quad (\text{WDB23})$$

The long exact sequences of the finite filtration (WDB17) and the block-triangular argument following (WDB3) now give

$$w_q(\alpha) \leq 2 + 2^{-k} \quad \text{on } E_2^{11} = H^1(P, \mathcal{R}^1). \quad (\text{WDB24})$$

For the two outer terms, 3.2.12 uses the elementary degree-zero and degree-two cohomology of curves from 1.4, not the theorem being proved. On any fibre, H_c^0 is a space of sections of the weight-zero lisse sheaf and its eigenvalues have weight zero. For H_c^2 , remove the finitely many singular points and pass to smooth connected components. Formula 1.4.1.3 identifies the top cohomology with geometric coinvariants of a stalk, followed by the twist (-1) ; its eigenvalues have weight two. If components are permuted, a finite extension makes them individually defined, and the weight identity (WDB1) descends the same bound. Hence the stalk weights of $\mathcal{R}_0^0$ and $\mathcal{R}_0^2$ are respectively at most 0 and 2.

Applying the same elementary curve statements on the base proves

$$w_q(\alpha) \leq 2 \quad \text{on } E_2^{02} = H^0(P, \mathcal{R}^2) \text{ and } E_2^{20} = H^2(P, \mathcal{R}^0). \quad (\text{WDB25})$$

For the first assertion, a nonzero global eigensection is nonzero at some closed-point stalk; Frobenius there acts by the corresponding power of its global eigenvalue. For the second, removal of finitely many points does not change H_c^2 ; on a dense lisse open, 1.4.1.3 expresses it as coinvariants of a stalk with weights at most zero, twisted by (-1) . This adds exactly two. These arguments also cover constructible sheaves on the base with singular boundary stalks. The β in the printed explanation of 3.2.12 is specialized here to the auxiliary weight zero of $\mathcal{G}_0$.

Equations (WDB24)–(WDB25) and the spectral sequence (WDB12) prove

$$w_q(\mu) \leq 2 + 2^{-k} \quad \text{for every eigenvalue on } H_c^2(\widetilde{V}, \pi^* \mathcal{G}). \quad (\text{WDB26})$$

Only one error 2^{-k} has appeared: it enters through the base cohomology in (WDB23). The fibre coefficient weight has already been reduced to at most one by the separate integer and strict-bound argument. Applying B_k twice without that reduction would instead give $2 + 2 \cdot 2^{-k}$ and no improvement after taking a square root.

The nonzero class survives the two exact injections

Künneth gives a Frobenius-equivariant direct-sum decomposition in degree two. Its $(1, 1)$ summand provides the first map below; the second is pullback along the blow-up:

$$\begin{aligned} H_c^1(U, \mathcal{F}^{[0]}) \otimes H_c^1(U, \mathcal{F}^{[0]}) &\hookrightarrow H_c^2(V, \mathcal{G}) \\ &\hookrightarrow H_c^2(\widetilde{V}, \pi^* \mathcal{G}). \end{aligned} \quad (\text{WDB27})$$

The second injection is 3.1.1.4. More precisely 3.1.1.3 identifies

$$H_c^2(\widetilde{V}, \pi^* \mathcal{G}) \simeq H_c^2(V, \mathcal{G}) \oplus H^0(V \cap A, \mathcal{G})(-1), \quad (\text{WDB28})$$

with pullback as the inclusion of the first summand. The center is disjoint from D by (B), so these are the actual blow-up centers in the open surface. Equivalently the Poincaré-dual transpose described in 3.1.1.4 supplies a retraction. The extra summand in (WDB28) is retained rather than identified with zero.

If $v \neq 0$ is an eigenvector of F with eigenvalue $\alpha^{[0]}$ in the original degree-one space, choose a linear functional λ with $\lambda(v) = 1$. Then $(\lambda \otimes \lambda)(v \otimes v) = 1$, so $v \otimes v \neq 0$. Both maps in (WDB27) are injective, hence their composite carries it to a nonzero eigenvector with eigenvalue $(\alpha^{[0]})^2$. No exterior square is being taken: the product is on two copies of the curve, so the odd cohomological degree does not force this tensor to vanish. Invertibility and the existence of an eigenvector hold over $\overline{\mathbb{Q}}_\ell$ without any diagonalizability assumption.

Applying (WDB26) to this specific image gives

$$\begin{aligned} 2(w_q(\alpha) - \beta) &= w_q((c\alpha)^2) \leq 2 + 2^{-k}, \\ w_q(\alpha) &\leq \beta + 1 + 2^{-(k+1)}. \end{aligned} \tag{WDB29}$$

This proves B_{k+1} under (A)–(C), with the original eigenvalue and original coefficient weight restored exactly.

Removing the reduction hypotheses preserves the offending eigenvalue

For a finite extension $\mathbb{F}_{q^d}/\mathbb{F}_q$, the geometric cohomology space is unchanged and F is replaced by F^d . Thus an eigenvalue α is replaced by α^d , and (WDB1) proves equality of its old and new weights. A bound over the extension gives the identical bound over the original field. Deligne uses this operation in 3.2.14 to define over a finite field an embedding and a general-position pencil that exist over the algebraic closure, and then to make their finitely many exceptional points rational. This removes (B) and (C). It does not itself divide any exponent error by d .

To remove (A), the monodromy theorem supplies a finite surjective cover $g : U'_0 \rightarrow U_0$, with U'_0 smooth, for which $g^*\mathcal{F}_0^{[0]}$ has tame unipotent boundary monodromy. Purity is preserved by pullback: if x' lies over x with residue degree e , its Frobenius eigenvalues are the e -th powers of those at x , and $N(x') = N(x)^e$. The exact map used to retain cohomology is

$$g^* : H_c^a(U, \mathcal{F}^{[0]}) \hookrightarrow H_c^a(U', g^*\mathcal{F}^{[0]}). \tag{WDB30}$$

Here is a trace construction which includes the multiplicities at ramification points. On a connected smooth target curve the finite surjective map is flat of constant degree $d_g > 0$: its finite local modules are torsion-free over the discrete valuation rings of the target and hence free. At a geometric point u , the trace of the constant coefficient sheaf $E = \overline{\mathbb{Q}}_\ell$ is

$$\mathrm{Tr}_{g,u} : \bigoplus_{u'|u} E \longrightarrow E, \quad (a_{u'}) \longmapsto \sum_{u'|u} \mathrm{length}(\mathcal{O}_{g^{-1}(u),u'}) a_{u'}. \tag{WDB30a}$$

The sum of these lengths is d_g . Compatibility with restriction follows by decomposing the inverse image of an étale neighborhood into open and closed pieces: each piece is finite flat and its rank is locally constant. Thus (WDB30a) is a sheaf morphism $g_*E \rightarrow E$, with the unit followed by trace equal to $d_g \mathrm{id}$. The projection formula tensors this identity with $\mathcal{F}^{[0]}$. Finite maps are proper and their geometric fibres have no higher étale cohomology, so $Rg_* = g_* = g_!$. Taking compact-support cohomology gives the map (WDB30) with retraction $d_g^{-1} \mathrm{Tr}_g$. The coefficient field has characteristic zero, so the inverse d_g^{-1} exists even if ℓ divides d_g . The formula includes inseparable lengths as well. Because these maps are Frobenius-equivariant, they retain the same eigenvalue. Connected components can be treated after a finite base extension and descended by (WDB1). The universal B_{k+1} follows.

A finite contradiction, followed by the exact dual lower bound

Induction now proves (WDB2) for every integer $k \geq 0$. Suppose an actual eigenvalue had $w_q(\alpha) = \beta + 1 + \varepsilon$ with $\varepsilon > 0$. The specific finite integer

$$K = \max\{0, \lfloor \log_2(1/\varepsilon) \rfloor + 1\} \quad \text{satisfies} \quad 2^{-K} < \varepsilon. \quad (\text{WDB31})$$

Indeed if $\varepsilon > 1$, $K = 0$ works; otherwise $K > \log_2(1/\varepsilon)$. At this finite stage (WDB2) would say $\beta + 1 + \varepsilon \leq \beta + 1 + 2^{-K}$, a contradiction. Consequently

$$w_q(\alpha) \leq \beta + 1 \quad \text{on } H_c^1(U, \mathcal{F}). \quad (\text{WDB32})$$

This is the dyadic 2^{-k} bootstrap in Weil II, not the $1/k$ error obtained from arbitrarily large products in the earlier proof of Weil I. Both retain the offending eigenvalue, but their geometric estimates and induction parameters differ.

Let now $j : U_0 \hookrightarrow X_0$ with X_0 smooth projective. Equation (WDB8) transfers (WDB32) to $H^1(X, j_*\mathcal{F})$. Deligne's duality in 3.2.15 is the perfect Frobenius-equivariant pairing with values in the trivial coefficient field

$$H^1(X, j_*\mathcal{F}) \otimes H^1(X, j_*\mathcal{F}^\vee(1)) \longrightarrow \overline{\mathbb{Q}}_\ell. \quad (\text{WDB33})$$

Its perfectness on these actual image groups can also be seen directly from Poincaré duality on U . The compact-to-ordinary maps for $\mathcal{F}$ and $\mathcal{F}^\vee(1)$ are transposes, up to the degree-one cup-product sign. The kernel of either transpose is the annihilator of the other image. Quotienting the compactly supported groups by these kernels therefore induces a nondegenerate pairing on the two images. Their identification with the two j_* cohomology groups gives (WDB33), including its scalar target. The dual coefficient sheaf has weight $-\beta - 2$. For an eigenvalue α of the first factor, α^{-1} occurs in the second, whether or not Frobenius is semisimple. Applying the already proved upper bound to that factor gives

$$-w_q(\alpha) = w_q(\alpha^{-1}) \leq (-\beta - 2) + 1 = -\beta - 1. \quad (\text{WDB34})$$

Together with the upper bound this proves $w_q(\alpha) = \beta + 1$. In degrees zero and two, the invariant and coinvariant descriptions of 1.4, with the Tate twist (-1) in degree two, give weights β and $\beta + 2$. Thus all three degrees give 3.2.3:

$$w_q(\alpha) = \beta + i \quad \text{on } H^i(X, j_*\mathcal{F}). \quad (\text{WDB35})$$

For degree one this group is also the image of $H_c^1(U, \mathcal{F}) \rightarrow H^1(U, \mathcal{F})$, as stated at the start of 3.2.3. Purity is being asserted for this actual image; the mixed boundary part of all of $H_c^1(U, \mathcal{F})$ has not been declared pure.

The mixed-sheaf theorem retains and controls the boundary

Theorem 3.3.1 concerns a morphism $f : X \rightarrow Y$ of schemes of finite type over $\mathbb{Z}$, with ℓ invertible as in Deligne's conventions. It asserts that a mixed constructible sheaf $\mathcal{F}$ of weights at most n has $R^i f_* \mathcal{F}$ mixed of weights at most $n + i$. Deligne proves it by reduction to the curve theorem together with the local boundary calculation. The reductions are by exact sequences in the sheaf; decomposition into an open set and its closed complement; base change on the target; and the Frobenius-equivariant Leray spectral sequence for a composition. For the sheaf reduction involving a kernel, the preceding term in the long exact sequence has bound $n + i - 1$, which is at most $n + i$; thus that reduction also preserves the claimed bound. Relative dimension zero contributes only degree zero and preserves the pointwise weight. Universal homeomorphisms preserve the *étale* theory. Stratification and composition reduce to relative curves and then to a lisse pure coefficient sheaf. These are the explicit reductions 3.3.1(a)–(f).

On the generic relative curve one removes finitely many singular points, takes a finite covering with tame ramification at infinity, and uses the trace splitting of its coefficient sheaf. A purely inseparable field extension is allowed in making the smooth compactification. Deligne then spreads this finite geometric data over a neighborhood of the generic point of the reduced target. Reductions 3.3.1(c),(e) make this neighborhood the whole reduced target and allow the smooth reduced target used in Deligne's relative boundary calculation; the target reduction handles its complement. The resulting relative geometry is

$$\begin{aligned} X &\xrightarrow{j} \overline{X} \xleftarrow{i} D, & f &= \bar{f}j, \\ \bar{f}: \overline{X} &\longrightarrow Y \text{ smooth projective of relative dimension 1,} \\ D &\longrightarrow Y \text{ finite étale.} \end{aligned} \tag{WDB36}$$

Here $\mathcal{F}$ is lisse pure of weight n on X and tamely ramified along D . There is an actual short exact sequence on the compactification

$$0 \rightarrow j_! \mathcal{F} \rightarrow j_* \mathcal{F} \rightarrow i_* B \rightarrow 0, \quad B = i^* j_* \mathcal{F}. \tag{WDB37}$$

The boundary stalk is inertia invariants, not the entire generic stalk. By 1.8.6–1.8.8 its induced monodromy filtration satisfies

$$\mathrm{Gr}_a^M B = 0 \ (a > 0), \quad \mathrm{Gr}_a^M B \text{ pure of weight } n + a \ (a \leq 0). \tag{WDB38}$$

The signs here follow from $V^I \subset \ker N \subset M_0 V$; when a finite residual inertia group remains, its invariant functor is exact in characteristic zero and retains the same upper indices. These sheaves and their filtrations commute with passage to the fibres in (WDB36).

Applying 3.2.3 on each fibre, and for every coefficient embedding ι , proves that $R^a \bar{f}_* j_* \mathcal{F}$ is pointwise pure of weight $n + a$. Because $D \rightarrow Y$ is finite, its higher direct images vanish and its direct image carries the finite filtration (WDB38) to weights at most n . The long exact sequence of (WDB37) contains, in degrees zero and one,

$$\begin{aligned} 0 \rightarrow f_! \mathcal{F} &\rightarrow \bar{f}_* j_* \mathcal{F} \rightarrow (\bar{f}i)_* B \rightarrow R^1 f_! \mathcal{F} \\ &\rightarrow R^1 \bar{f}_* j_* \mathcal{F} \rightarrow 0, \end{aligned} \tag{WDB39}$$

and in degrees $a \geq 2$ gives $R^a f_! \mathcal{F} \simeq R^a \bar{f}_* j_* \mathcal{F}$. Thus degree zero has weights at most n ; degree one is an extension of the pure sheaf $R^1 \bar{f}_* j_* \mathcal{F}$, of weight $n + 1$, by the boundary quotient $\mathrm{coker}(\bar{f}_* j_* \mathcal{F} \rightarrow (\bar{f}i)_* B)$, of weights at most n ; and degree two has weight $n + 2$. This proves the desired upper bound in the reduced relative-curve case, and the exact reductions give 3.3.1. Boundary contributions have been included through (WDB37)–(WDB39), with their weights and maps specified.

For a finite-type $\mathbb{F}_q$ -scheme this gives 3.3.4: $H_c^i(X, \mathcal{F})$ has weights at most $n + i$ when $\mathcal{F}_0$ is mixed of weights at most n . If in addition X_0 is smooth of pure dimension d and $\mathcal{F}_0$ is lisse and mixed of weights at least n , then $\mathcal{F}_0^\vee$ has weights at most $-n$. Apply the upper bound to the exact Poincaré dual:

$$H^i(X, \mathcal{F})^\vee \simeq H_c^{2d-i}(X, \mathcal{F}^\vee)(d), \quad -n + (2d - i) - 2d = -n - i. \tag{WDB40}$$

Duality reverses weights, giving the lower bound $n + i$ for $H^i(X, \mathcal{F})$, which is 3.3.5. For a lisse pure sheaf of weight n , the natural map from compactly supported to ordinary cohomology is Frobenius-equivariant, and its image is both a quotient of a space of weights at most $n + i$ and a subspace of one of weights at least $n + i$. The characteristic-polynomial argument after (WDB3) then proves the exact conclusion 3.3.6:

$$\mathrm{im}(H_c^i(X, \mathcal{F}) \longrightarrow H^i(X, \mathcal{F})) \text{ is pure of weight } n + i. \tag{WDB41}$$

The original compact-support and ordinary cohomology groups are preserved in this statement; purity is proved for their specified common image.

The exact contradiction mechanism available for comparison

The decisive numerical chain in this proof is

$$\begin{aligned}
 & \text{nonzero local boundary quotient} \xrightarrow{\text{(WDB14)–(WDB19)}} b_a \in \mathbb{Z}, \\
 & \{b_a \in \mathbb{Z}, b_a < 2\} \implies b_a \leq 1, \\
 & \text{all local boundary quotients zero} \xrightarrow{\text{(WDB18)–(WDB22)}} H^1(P, \mathcal{K}_a) = 0, \tag{WDB42} \\
 & \mathbf{B}_k \xrightarrow{\text{(WDB23)–(WDB26)}} w_q(\mu) \leq 2 + 2^{-k}, \\
 & \alpha^{[0]} \xrightarrow{\text{(WDB27)}} 7 \implies (\alpha^{[0]})^2 \implies \mathbf{B}_{k+1}.
 \end{aligned}$$

The word “boundary” in this chain names the actual specialization-cokernel map (WDB18) into vanishing cycles, and the constant alternative has the actual cohomology-vanishing map (WDB22). The improvement is proved only after both alternatives have been treated. The finite integer (WDB31) then excludes every strict upper violation, and (WDB33)–(WDB34) exclude every strict lower violation of the pure image weight. Equations (WDB37)–(WDB41) extend this calculation to mixed coefficients without deleting their boundary-supported pieces.